TECHNICAL REPORT

Carbon Capture and Utilization (CCU): A Modular Closed-Loop Carbon Management Platform for Sustainable Cement Manufacturing

Executive Summary

Climate change has emerged as one of the greatest environmental challenges of the twenty-first century. Increasing concentrations of greenhouse gases, particularly carbon dioxide (CO_2), have accelerated global warming, resulting in rising temperatures, melting glaciers, sea-level rise, and an increase in the frequency of extreme weather events. While renewable energy and electrification have helped reduce emissions in several sectors, certain industries remain inherently difficult to decarbonize. Among these, the cement industry is one of the largest contributors to global carbon emissions due to the chemical processes involved in cement manufacturing.

This report proposes a **Modular Closed-Loop Carbon Capture and Utilization (CCU) System** designed specifically for the cement industry. Unlike conventional carbon capture technologies that primarily focus on storing captured carbon dioxide underground, this concept captures CO_2 directly from cement plant exhaust gases, purifies and compresses it, and immediately reuses it within the same industrial ecosystem to manufacture carbon-negative concrete. The objective is to convert carbon dioxide from an industrial waste product into a valuable raw material while significantly reducing the net carbon footprint of cement manufacturing.

The proposed system is based on a modular architecture that allows it to be retrofitted onto existing cement plants without requiring complete redesign of industrial infrastructure. Captured carbon dioxide is chemically mineralized within fresh concrete, forming stable calcium carbonate crystals that permanently lock the carbon inside the concrete while simultaneously improving its mechanical properties.

This report explores the scientific principles behind Carbon Capture and Utilization, the chemistry involved in carbon mineralization, the engineering considerations required for implementation, and the potential environmental and economic benefits of integrating such a system into modern cement manufacturing facilities.

1. Introduction

Industrial development has transformed modern civilization, enabling rapid urbanization, technological advancement, and improved standards of living. However, this progress has also resulted in a dramatic increase in greenhouse gas emissions, particularly carbon dioxide (CO_2). According to international climate assessments, atmospheric carbon dioxide concentrations have increased from approximately **280 ppm during the pre-industrial era to well above 420 ppm today**, making CO_2 the dominant anthropogenic greenhouse gas responsible for global warming.

Among industrial sectors, cement manufacturing plays a unique role. Unlike electricity generation, where renewable energy can gradually replace fossil fuels, cement production generates carbon dioxide through an unavoidable chemical reaction known as **calcination**. This means that even if cement plants were powered entirely by renewable electricity, a significant portion of their emissions would still remain.

As urban populations continue to grow, the demand for infrastructure—including roads, bridges, buildings, hospitals, airports, and housing—is expected to increase substantially. Since concrete is the most widely used construction material in the world, global cement production is projected to remain high for decades. Consequently, reducing emissions from the cement industry has become a priority in achieving international climate goals.

Carbon Capture and Utilization (CCU) offers a practical solution to this challenge by transforming emitted carbon dioxide into a useful industrial resource rather than allowing it to enter the atmosphere. Instead of treating CO_2 solely as waste, CCU enables industries to capture, process, and reuse carbon dioxide in various applications such as concrete production, synthetic fuels, chemicals, and industrial manufacturing.

The innovation proposed in this report focuses on integrating carbon capture directly with concrete production, creating a circular industrial process in which carbon emissions become an input for manufacturing carbon-negative concrete.

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2. Understanding Climate Change

Climate change refers to long-term changes in global temperature and weather patterns primarily caused by increased concentrations of greenhouse gases in Earth's atmosphere.

Greenhouse gases absorb infrared radiation emitted by Earth's surface and re-radiate it back toward the planet. This natural process, known as the **greenhouse effect**, maintains Earth's average temperature at approximately **15°C**, making life possible. However, excessive greenhouse gas emissions have intensified this effect, leading to an enhanced greenhouse effect that causes global warming.

The primary greenhouse gases include:

Gas	Approximate Contribution
Carbon Dioxide (CO_2)	Largest contributor
Methane (CH_4)	High warming potential
Nitrous Oxide (N_2O)	Long atmospheric lifetime
Water Vapour	Natural greenhouse gas
Fluorinated Gases	Extremely high warming potential

Among these gases, carbon dioxide is responsible for the majority of human-induced warming because of the enormous quantities released through industrial activity and fossil fuel combustion.

Major sources of carbon dioxide emissions include:

- Electricity generation
- Cement manufacturing
- Steel production
- Transportation
- Oil refining
- Chemical industries
- Deforestation

According to international energy reports, industrial activities account for approximately one-quarter of global greenhouse gas emissions, making industrial decarbonization one of the most important challenges of modern engineering.

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3. Why the Cement Industry?

The cement industry is one of the most difficult sectors to decarbonize because carbon dioxide emissions arise from two fundamentally different sources.

The first source is **fuel combustion**, where fossil fuels such as coal, petroleum coke, or natural gas are burned to generate temperatures of approximately **1450°C** inside rotary kilns.

The combustion reaction is represented by:

$$\mathbf{C + O_2 \rightarrow CO_2 + Heat}$$

This reaction provides the thermal energy required to manufacture clinker, the primary ingredient in cement.

The second source, and the more significant one, is **calcination**, a chemical decomposition reaction that occurs when limestone is heated to high temperatures.

The balanced chemical equation is:

$$\mathbf{CaCO_3 \rightarrow CaO + CO_2}$$

Where:

- $CaCO_3$ = Calcium Carbonate (Limestone)
- CaO = Calcium Oxide (Quicklime)
- CO_2 = Carbon Dioxide

This reaction is unavoidable because calcium oxide is the essential ingredient required to manufacture cement clinker.

Approximately **60% of cement plant emissions originate from calcination**, while the remaining **40% originate from fuel combustion**. This distinction is extremely important because replacing fossil fuels alone cannot eliminate emissions from cement manufacturing.

As a result, carbon capture technologies have become one of the few viable solutions capable of significantly reducing emissions from the cement sector.

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4. Why Carbon Capture and Utilization (CCU)?

Carbon Capture and Utilization is a technology that captures carbon dioxide before it is released into the atmosphere and converts it into useful products instead of treating it as waste.

Unlike traditional Carbon Capture and Storage (CCS), which permanently stores compressed carbon dioxide underground, CCU focuses on creating economic value from captured carbon.

Common applications include:

- Carbon-negative concrete
- Synthetic aviation fuels
- Methanol production
- Carbonates
- Industrial chemicals
- Dry ice
- Food-grade carbon dioxide
- Green building materials

Among these applications, carbon-negative concrete is considered one of the most promising because the carbon dioxide becomes chemically mineralized and permanently locked inside the concrete structure.

This creates a permanent carbon sink while simultaneously improving concrete durability and compressive strength.

End of Part 1.

Part 2 will go much deeper into **cement manufacturing**, **kiln operation**, **the complete chemistry of clinker formation**, and **every chemical reaction involved before we begin the CCU process itself**. That section is where the report starts becoming technically substantial.

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PART 2

5. Cement Manufacturing Process

To understand how Carbon Capture and Utilization (CCU) can reduce emissions, it is first necessary to understand how cement is manufactured and why this process produces such large quantities of carbon dioxide.

Cement production is one of the most energy-intensive industrial processes in the world. Modern cement plants operate continuously, producing thousands of tonnes of cement every day. The manufacturing process consists of several interconnected stages, each contributing to the quality of the final product.

Stage 1: Limestone Quarrying

The primary raw material used in cement manufacturing is **limestone ($CaCO_3$)**, which typically constitutes about 75–80% of the raw mix. Other materials such as clay, shale, sand, bauxite, and iron ore are added to provide silica (SiO_2), alumina (Al_2O_3), and iron oxide (Fe_2O_3), which are required for clinker formation.

Large rocks are extracted using controlled blasting techniques and transported to the crushing plant.

Typical raw material composition:

Material	**Purpose**
Limestone	Source of Calcium
Clay	Source of Silica and Alumina
Sand	Additional Silica
Iron Ore	Iron Oxide
Bauxite	Alumina

Stage 2: Crushing

The quarried limestone can exceed one metre in diameter. Before further processing, it must be reduced to smaller pieces using jaw crushers, hammer crushers, or impact crushers.

The crushed material is typically reduced to particles less than **25 mm** in size.

This stage consumes electrical energy but produces negligible carbon emissions compared to the kiln.

Stage 3: Raw Meal Preparation

The crushed limestone and other raw materials are transported to a raw mill where they are finely ground into a powder known as **raw meal**.

The particle size of the raw meal is extremely important because uniform particles improve heat transfer inside the kiln and ensure complete chemical reactions.

Modern cement plants continuously analyse the chemical composition of the raw meal using X-ray fluorescence (XRF) analysers to maintain consistency.

Typical composition:

- Calcium Oxide (CaO)
- Silicon Dioxide (SiO_2)
- Aluminium Oxide (Al_2O_3)
- Iron Oxide (Fe_2O_3)

Stage 4: Preheater Tower

Before entering the rotary kiln, the raw meal passes through a series of cyclone preheaters.

The temperature gradually increases from approximately **100°C to 900°C**.

The objectives of preheating are:

- Remove moisture
- Reduce fuel consumption
- Improve kiln efficiency
- Begin decomposition reactions

At temperatures around **700–900°C**, limestone starts decomposing.

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6. Calcination

Calcination is the single most important reaction in cement manufacturing and the largest source of industrial carbon dioxide emissions.

During calcination, calcium carbonate decomposes into calcium oxide and carbon dioxide.

Chemical Equation

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

This reaction begins around **825°C** and becomes extensive above **900°C**.

Why this reaction matters

Calcium oxide is essential for clinker formation.

Without this reaction:

- Cement cannot be manufactured.
- No substitute industrial process currently eliminates this chemistry.

This is why cement is classified as a **hard-to-abate industry**.

Approximately **0.53 tonnes of CO_2** are released purely from calcination for every tonne of clinker produced.

This carbon dioxide is called **process CO_2** because it originates from the raw material itself rather than fuel combustion.

Energy Requirement

Calcination is highly endothermic.

Approximately

178 kJ/mol

of heat is required.

This is one reason cement plants consume enormous amounts of fuel.

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7. Rotary Kiln

The rotary kiln is the heart of every cement plant.

Typical specifications:

Length:
40–70 metres

Diameter:
4–7 metres

Temperature:
1450°C

Residence time:
20–30 minutes

The kiln slowly rotates while the raw meal moves downward.

Inside the kiln several reactions occur simultaneously.

Formation of Clinker Minerals

After calcination, calcium oxide reacts with silica, alumina and iron oxide to produce clinker.

Major clinker minerals include

Tricalcium Silicate (C_3S)

$3CaO + SiO_2 \rightarrow 3CaO \cdot SiO_2$

This compound provides early strength to concrete.

Dicalcium Silicate (C_2S)

$2CaO + SiO_2 \rightarrow 2CaO \cdot SiO_2$

Responsible for long-term strength.

Tricalcium Aluminate (C_3A)

$3CaO + Al_2O_3 \rightarrow 3CaO \cdot Al_2O_3$

Responsible for rapid hydration.

Tetracalcium Aluminoferrite (C_4AF)

$4CaO + Al_2O_3 + Fe_2O_3 \rightarrow$

$4CaO \cdot Al_2O_3 \cdot Fe_2O_3$

Improves manufacturing properties.

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8. Cooling and Grinding

The hot clinker leaving the kiln is cooled rapidly.

Rapid cooling improves clinker quality and allows heat recovery.

The cooled clinker is then mixed with approximately **3–5% gypsum ($CaSO_4 \cdot 2H_2O$)**.

Gypsum prevents flash setting during cement hydration.

Finally, the mixture is ground into fine cement powder.

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9. Sources of CO_2 Emissions in Cement Plants

Understanding the origin of emissions helps determine where carbon capture systems should be installed.

Source	Approximate Contribution
Calcination	~60%
Fuel Combustion	~40%

This distinction is important because switching to renewable electricity only addresses combustion-related emissions. Calcination emissions remain unavoidable without carbon capture.

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10. Flue Gas Composition

The gases leaving the kiln are collectively called **flue gas**.

A typical flue gas composition from a cement plant is:

Component	**Typical Range**
Nitrogen (N_2)	60–70%
Carbon Dioxide (CO_2)	18–25%
Water Vapour (H_2O)	5–10%
Oxygen (O_2)	2–5%
Sulphur Oxides (SO_x)	Trace
Nitrogen Oxides (NO_x)	Trace
Dust/Particulates	Variable

The relatively high concentration of CO_2 in cement plant exhaust makes it well suited for **point-source carbon capture**, which is generally more energy-efficient than Direct Air Capture (DAC).

Why Point-Source Capture?

Capturing CO_2 directly from industrial exhaust offers several advantages:

- Higher CO_2 concentration than atmospheric air, making separation easier.
- Lower energy consumption compared to Direct Air Capture.
- Existing exhaust ducts can be retrofitted with capture systems.
- Continuous operation provides a steady supply of CO_2 for utilization.
- Greater economic viability for large industrial facilities.

Because of these advantages, the proposed system focuses on capturing CO_2 directly from cement plant flue gases before they are released into the atmosphere.

Transition to the CCU Process

Once the flue gas leaves the cement plant, it cannot be sent directly to a carbon utilization system. It must first undergo several stages of conditioning to remove dust, cool the gas, separate carbon dioxide from other gases, regenerate the capture solvent, and compress the purified CO_2 for utilization.

The following section explains each stage of the Carbon Capture and Utilization process in detail, including the underlying chemistry, engineering principles, and the rationale behind each step.

End of Part 2

TECHNICAL REPORT

Carbon Capture and Utilization (CCU): A Modular Closed-Loop Carbon Management Platform for Sustainable Cement Manufacturing

PART 3

11. The Complete Carbon Capture and Utilization (CCU) Process

Once carbon dioxide is generated inside the cement kiln, the objective is to prevent it from being released into the atmosphere. Carbon Capture and Utilization (CCU) achieves this by intercepting the exhaust gas stream, separating carbon dioxide from the other gases, purifying it, compressing it, and converting it into a useful industrial product.

The proposed system follows eight major stages:

1. Flue Gas Collection
2. Gas Pre-treatment
3. Carbon Capture
4. Solvent Regeneration
5. Carbon Dioxide Purification
6. Compression
7. Temporary Storage
8. Utilization in Carbon-Negative Concrete

Each stage plays a crucial role in ensuring that the captured CO_2 is of sufficient purity for industrial reuse.

Stage 1 – Flue Gas Collection

After clinker formation, the exhaust gases exit the rotary kiln through a network of ducts. These gases are extremely hot, typically between **250°C and 400°C**, and contain not only carbon dioxide but also dust, nitrogen, oxygen, water vapour, sulfur oxides, and nitrogen oxides.

Typical flue gas composition:

Component	Percentage
Nitrogen (N_2)	60–70%
Carbon Dioxide (CO_2)	18–25%
Water Vapour (H_2O)	5–10%
Oxygen (O_2)	2–5%
Sulfur Oxides (SO_x)	<1%
Nitrogen Oxides (NO_x)	<1%
Dust	Variable

At this stage, the gas cannot be captured directly because impurities would damage the capture equipment and reduce efficiency.

Stage 2 – Gas Pre-Treatment

Before carbon dioxide can be separated, the flue gas undergoes cleaning and conditioning.

Dust Removal

Cement exhaust contains significant quantities of fine cement particles. These particles can clog equipment and contaminate solvents.

Industries typically use one or more of the following systems:

Cyclone Separators

Cyclone separators remove heavier dust particles by using centrifugal force.

Advantages:

- Simple construction
- Low maintenance
- No moving parts
- Handles high temperatures

However, cyclone separators cannot remove very fine particles.

Bag Filters

The partially cleaned gas passes through fabric filters.

These filters capture particles as small as **1 micrometre**.

Advantages:

- High efficiency (>99%)
- Relatively simple maintenance
- Low operating cost

Electrostatic Precipitators (ESP)

Large cement plants often use Electrostatic Precipitators.

Working principle:

1. Gas enters the ESP.
2. High-voltage electrodes ionize the gas.
3. Dust particles become electrically charged.
4. Charged particles are attracted to metal collection plates.
5. The collected dust is periodically removed.

Collection efficiencies greater than **99.5%** are common.

Gas Cooling

After particulate removal, the gas is cooled to approximately **40–60°C**.

Cooling is essential because the chemical solvent used for CO_2 capture performs best at moderate temperatures.

Gas cooling is usually achieved using:

- Heat exchangers
- Water spray towers
- Cooling towers

Cooling also condenses excess water vapour, reducing the load on downstream equipment.

Stage 3 – Carbon Capture

The most commercially mature method for post-combustion carbon capture is **amine absorption**.

This method is widely used because it offers:

- High CO_2 capture efficiency
- Proven industrial performance
- Compatibility with existing cement plants

What Are Amines?

Amines are nitrogen-containing organic compounds that chemically react with carbon dioxide.

The most common solvent is:

Monoethanolamine (MEA)

Other industrial solvents include:

- Diethanolamine (DEA)
- Methyldiethanolamine (MDEA)
- Piperazine blends

Absorber Column

The cooled flue gas enters the bottom of a tall absorber column.

MEA solution flows downward from the top.

As the two streams move in opposite directions, carbon dioxide reacts with the amine solution while nitrogen, oxygen, and other gases continue upward and exit the stack.

This counter-current flow maximizes contact time and improves capture efficiency.

Chemical Reaction

The simplified reaction between carbon dioxide and monoethanolamine is:

$$\mathbf{CO_2 + 2RNH_2 \rightleftharpoons RNHCOO^- + RNH_3^+}$$

Where:

- CO_2 = Carbon dioxide
- RNH_2 = Amine molecule
- $RNHCOO^-$ = Carbamate ion
- RNH_3^+ = Protonated amine

This reaction is reversible, allowing the solvent to be regenerated and reused.

Typical industrial capture efficiencies range from **85–95%**.

Why Use MEA?

Advantages:

- High reaction rate with CO_2
- Well-established technology
- Suitable for retrofitting existing plants
- High capture efficiency

Limitations:

- Requires heat for regeneration
- Solvent degradation over time
- Corrosion if not properly managed

Stage 4 – Solvent Regeneration

Once the amine solution has absorbed CO_2, it becomes known as **rich amine**.

The rich amine is pumped into a second vessel called the **stripper** or **regeneration column**.

Steam heats the solution to approximately **100–120°C**.

The added heat reverses the absorption reaction:

$$\mathbf{RNHCOO^- + Heat \rightarrow CO_2 + 2RNH_2}$$

This releases nearly pure carbon dioxide gas while regenerating the amine solvent.

The regenerated **lean amine** is cooled and pumped back to the absorber, creating a continuous closed-loop process.

Why Regeneration Matters

Without regeneration:

- New solvent would constantly be required.
- Operating costs would be extremely high.
- Large-scale operation would not be economically viable.

By recycling the solvent, operating costs are significantly reduced.

Stage 5 – Carbon Dioxide Purification

The gas leaving the stripper contains mainly carbon dioxide but may still contain:

- Water vapour

- Trace solvent vapours
- Oxygen
- Nitrogen

To prepare it for industrial use, it undergoes further purification.

Typical purification steps include:

- Moisture removal using dryers
- Activated carbon filters
- Fine particulate filters
- Gas polishing units

The resulting CO_2 purity is typically **95–99%**, depending on the intended application.

Stage 6 – CO_2 Compression

Freshly separated carbon dioxide occupies a large volume, making transportation and storage impractical.

Compression reduces its volume dramatically.

Industrial systems usually use **multi-stage compressors**, where the gas is compressed in several steps with intercooling between stages to improve efficiency and reduce equipment stress.

Typical storage pressures range from **80–150 bar**, depending on the intended application.

Benefits of compression:

- Easier storage
- Reduced transportation costs
- Stable flow to downstream utilization systems
- Improved process efficiency

In the proposed system, compressed CO_2 is stored only briefly before being sent directly to the concrete production unit, minimizing long-term storage requirements.

Stage 7 – Temporary Storage

Unlike conventional Carbon Capture and Storage (CCS), which requires permanent geological storage, the proposed CCU system only needs short-term buffer storage.

The storage tank serves to:

- Balance fluctuations in CO_2 production.
- Ensure a steady supply for concrete manufacturing.
- Allow maintenance without interrupting the entire process.

Safety systems include:

- Pressure relief valves
- Gas leak detectors
- Temperature monitoring
- Emergency venting systems

Stage 8 – Carbon Utilization Through Mineralization

The final and most important stage of the process is **mineralization**, where captured carbon dioxide is permanently converted into a stable mineral inside fresh concrete.

During cement hydration, calcium hydroxide is naturally produced:

$$\mathbf{CaO + H_2O \rightarrow Ca(OH)_2}$$

The injected carbon dioxide reacts with calcium hydroxide:

$$\mathbf{CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O}$$

Another way to represent the precipitation step is:

$$\mathbf{Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3(s)}$$

The product, **calcium carbonate ($CaCO_3$)**, forms microscopic crystals that become permanently embedded within the concrete matrix.

This process:

- Permanently stores carbon.

- Increases concrete density.
- Improves compressive strength.
- Reduces porosity.
- Enhances durability.

Because the carbon is chemically locked into a stable mineral, it cannot easily re-enter the atmosphere under normal conditions.

Why This Matters

Traditional carbon capture often ends with compressed CO_2 being transported and stored underground. While effective, this approach generates little direct economic value.

The proposed **Modular Closed-Loop CCU System** instead transforms carbon dioxide from an industrial waste product into a valuable manufacturing input.

This creates a circular carbon flow:

Cement Production → CO_2 Capture → Purification → Compression → Concrete Production → Permanent Carbon Storage

By integrating carbon capture with concrete manufacturing on the same site, the system reduces transport requirements, creates value from emissions, and supports the production of lower-carbon construction materials.

End of Part 3

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PART 4

12. Proposed Innovation: A Modular Closed-Loop Carbon Management Platform

Introduction

While Carbon Capture and Utilization (CCU) is an established field, the proposed innovation focuses on **integrating multiple existing technologies into a single modular system** that can be retrofitted onto cement plants and directly connected to concrete production.

The objective is to create a **closed-loop industrial ecosystem**, where carbon dioxide emitted during cement manufacturing is captured, purified, compressed, and immediately reused within the same facility to manufacture carbon-negative concrete.

Instead of treating carbon dioxide as waste requiring disposal, the proposed system treats CO_2 as a valuable industrial raw material.

This approach reduces transportation requirements, lowers operational complexity, and supports the transition toward lower-emission cement manufacturing.

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13. System Architecture

The proposed system consists of six primary modules.

Module 1 – Flue Gas Collection

Purpose:

Collect exhaust gases directly from the cement kiln before they enter the atmosphere.

Major Components

- High-temperature ducting
- Flow control valves
- Temperature sensors
- Pressure sensors

Module 2 – Gas Cleaning

Purpose

Prepare the gas for carbon capture.

Equipment

- Cyclone separator

- Electrostatic precipitator
- Bag filter
- Cooling tower
- Moisture separator

Output

A clean flue gas stream suitable for carbon capture.

Module 3 – Carbon Capture

Purpose

Separate carbon dioxide from the remaining gases.

Major Components

- Absorber column
- MEA storage tank
- Solvent circulation pumps
- Heat exchanger
- Regeneration column

Output

High-purity carbon dioxide.

Module 4 – CO_2 Compression

Purpose

Reduce gas volume for efficient storage and utilization.

Components

- Multi-stage compressor
- Moisture removal unit

- Pressure regulators
- Storage cylinders

Output

Compressed CO_2.

Module 5 – Temporary Storage

Purpose

Provide continuous CO_2 supply to the concrete plant.

Storage System

- High-pressure storage vessels
- Safety valves
- Gas monitoring sensors

Unlike CCS systems, this storage is only temporary because the gas will be reused almost immediately.

Module 6 – Concrete Mineralization

Purpose

Inject purified carbon dioxide into fresh concrete.

Major Components

- Injection manifold
- Automated dosing system
- Mixing chamber
- Quality monitoring sensors

Output

Carbon-negative concrete.

14. Closed-Loop Carbon Cycle

The proposed system follows the cycle below:

Limestone

↓

Cement Manufacturing

↓

CO_2 Produced

↓

Gas Cleaning

↓

Carbon Capture

↓

CO_2 Purification

↓

Compression

↓

Temporary Storage

↓

Concrete Production

↓

Mineralization

↓

Carbon-Negative Concrete

Instead of releasing carbon dioxide into the atmosphere, the factory continually recycles it into a useful construction material.

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15. Why Make the System Modular?

Many existing cement plants have already invested millions of rupees in their production infrastructure.

Replacing an entire plant is economically unrealistic.

A modular design allows industries to install the system without rebuilding the factory.

Advantages include:

- Easier installation
- Lower initial investment
- Reduced downtime
- Easier maintenance
- Scalability
- Future upgrades without replacing the entire system

For example, a plant could begin with one capture unit and later add additional modules as production increases.

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16. Advantages Over Conventional CCS

Conventional CCS	**Proposed Modular CCU**
Stores CO_2 underground	Converts CO_2 into a useful product
Generates little direct revenue	Creates value through carbon-negative concrete
Requires pipelines and storage sites	Utilizes CO_2 on-site
Long-term monitoring required	Carbon stored within concrete
Higher transport costs	Minimal transport needed

17. Business Model

The success of any engineering solution depends not only on its technical performance but also on its commercial viability. The proposed Modular Closed-Loop CCU Platform is designed as a **Business-to-Business (B2B)** solution that targets industries seeking to reduce emissions while maintaining productivity.

Rather than selling carbon dioxide, the company would provide an integrated decarbonization platform combining equipment, engineering services, monitoring, and long-term operational support.

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18. Value Proposition

The proposed system creates value for customers in several ways:

Environmental Value

- Reduced industrial CO_2 emissions
- Permanent carbon storage
- Lower carbon footprint
- Contribution toward corporate sustainability goals

Economic Value

- Potential reduction in carbon compliance costs (where applicable)
- Higher-value low-carbon concrete products
- Lower transport costs through on-site utilization
- Improved resource efficiency

Operational Value

- Retrofit compatibility
- Modular expansion
- Continuous operation
- Minimal disruption to existing production

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19. Customer Segments

Primary Customers

- Cement manufacturers
- Integrated cement and concrete plants

- Ready-mix concrete producers
- Large infrastructure companies

Secondary Customers

- Construction firms
- Government infrastructure agencies
- Green building developers
- Industrial parks

Future Markets

- Steel plants
- Lime production facilities
- Chemical industries

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20. Revenue Streams

The business model relies on multiple revenue sources rather than a single product sale.

Equipment Sales

Revenue from supplying modular CCU units.

Installation Services

Engineering design, site preparation, installation, and commissioning.

Maintenance Contracts

Annual maintenance agreements covering inspections, spare parts, solvent management, and software updates.

Software & Monitoring

A digital dashboard can monitor:

- CO_2 captured
- CO_2 utilized
- System efficiency
- Energy consumption
- Emissions reduction
- Maintenance schedules

This could be offered as a subscription service.

Engineering Consulting

Support for plant integration, optimization, operator training, and regulatory compliance.

Carbon Accounting

The platform could generate verified reports on captured and utilized CO_2 to assist customers with sustainability reporting and environmental disclosures.

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21. Cost Structure

The primary costs of operating the business include:

Capital Costs

- Manufacturing equipment
- Compressors
- Absorber columns
- Pumps
- Control systems

Operating Costs

- Electricity
- Solvent replacement
- Routine maintenance
- Labour
- Transportation of equipment
- Research and development

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22. Key Partnerships

Potential strategic partners include:

- Cement manufacturers
- Concrete producers
- Engineering procurement and construction (EPC) companies
- Universities and research institutions
- Government agencies promoting industrial decarbonization
- Environmental certification organizations

These partnerships can support pilot projects, technical validation, and future commercialization.

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23. Implementation Roadmap

Phase 1 – Research and Design

- Finalize system design
- Simulate process performance
- Select capture technology
- Conduct laboratory validation

Phase 2 – Pilot Prototype

- Install a small-scale demonstration unit
- Test capture efficiency
- Evaluate concrete quality
- Optimize operating conditions

Phase 3 – Industrial Pilot

- Deploy the system at a partner cement facility
- Collect operational data
- Assess economic performance
- Refine system design

Phase 4 – Commercial Deployment

- Offer modular installations to industrial customers
- Expand service and maintenance operations
- Develop software and monitoring solutions

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24. Conclusion

The proposed Modular Closed-Loop Carbon Capture and Utilization Platform demonstrates how existing carbon capture technologies can be integrated into a practical industrial solution for the cement sector. By capturing carbon dioxide at its source, purifying it, compressing it, and reusing it within the same facility to produce carbon-negative concrete, the system transforms an unavoidable process emission into a valuable manufacturing input.

Although challenges remain—including capital costs, energy requirements, and large-scale implementation—the modular approach offers flexibility for retrofitting existing plants and scaling deployment over time. As governments, industries, and investors increasingly focus on decarbonization, solutions that combine environmental benefits with commercial value are likely to play an important role in the transition to a more sustainable construction industry.
